BUSINESS REQUIREMENTS DOCUMENT

Camera-Based Attendance & Workforce Tracking System

Multi-role enterprise platform for face-based attendance capture, identification, reporting, and workflow management

Prepared for

Omran

Prepared by

Muscat Tech Solutions

Document Version 1.0

Status: Draft — pending stakeholder review

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1. Document Overview

1.1 Purpose

This Business Requirements Document (BRD) describes the functional and non-functional requirements for a camera-based employee attendance system to be delivered for an enterprise client in Oman. The system performs face-based detection and identification from IP cameras on the corporate LAN, matches detections against an enrolled employee database, and produces attendance records, reports, and management workflows.

The document is based on an initial requirements discussion with the client, followed by a structured clarification exchange. It is intended to be reviewed and signed off by the client before implementation begins.

1.2 Scope

The system covers:

- Face detection and identification from IP cameras (RTSP) on the corporate network
- Employee enrollment via Excel import and reference-photo folder ingestion
- Attendance record generation with configurable time policies (fixed, flex, Ramadan, special schedules)
- Role-based dashboards for Admin, Manager, Employee, and HR
- Request/approval workflows for attendance exceptions (late-in, early-out, special absences)
- Automated daily reports delivered to HR and IT departments
- Camera health and capture-volume monitoring
- Microsoft Entra ID (Azure AD) authentication with email/password fallback for development
- Deployment on a single Ubuntu host on the corporate LAN, with optional future cloud deployment for read-only report access

1.3 Out of Scope (Phase 1)

The following items have been discussed but are explicitly deferred:

- Native iOS/Android mobile apps (responsive web UI is sufficient)
- ERP integration for report sync — architectural hook will be included, the actual connector is a future phase

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- Live camera feed viewable from the cloud (only derived data will sync to cloud, never raw RTSP)
- Face-based search (upload a photo to find who it is) — only text-based search in Phase 1
- Biometric-data-specific legal review — the system will include reasonable controls, but the client remains responsible for Oman PDPL compliance sign-off

1.4 Intended Audience

- Client stakeholders (HR, IT, Operations)
- Engineering team (backend, frontend, DevOps)
- QA and deployment engineers
- Project management and delivery team

1.5 Document Conventions

- Requirements are grouped by functional area, not by client interview order.
- Each requirement has a unique ID (e.g. FR-AUTH-01) for traceability.
- Items marked 'Assumption' reflect engineering interpretation where the client brief was ambiguous — these need explicit client confirmation.
- Items marked 'Open Question' are deferred to a later clarification round.

2. Business Context

2.1 Current Situation

The client currently relies on manual or badge-based attendance tracking. They have IP cameras (CP Plus) installed at entry/exit points and wish to leverage these for automated employee attendance capture. They have an existing HR process with defined shift policies and an approval chain (Manager → HR) for attendance exceptions.

2.2 Business Objectives

- Eliminate manual attendance tracking and badge-based systems
- Provide accurate, auditable attendance records tied to identified employees
- Give managers real-time visibility of their team's attendance
- Automate the daily reporting process to HR and IT
- Enable a formal request/approval workflow for attendance exceptions
- Remain aligned with Omani data protection obligations (PDPL, 2023)

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2.3 Key Success Criteria

- > 95% correct face identification for enrolled employees under normal camera conditions (front-facing, reasonable lighting)
- Daily attendance reports delivered automatically to HR and IT by configured time, with no manual intervention
- Managers can view their team's attendance and approve/reject requests within the system, without contacting IT
- Admin and HR can configure shift policies and holidays without engineering involvement
- System is accessible from any device on the corporate network via a browser

2.4 Stakeholders

Stakeholder	Interest
Client Admin / IT team	System configuration, user management, camera operations, infrastructure
HR department	Attendance reports, approval workflows, policy configuration, export to ERP
Line managers	Team attendance visibility, approval of exception requests
Employees	Personal attendance visibility, ability to submit request for exceptions
Engineering / delivery team	Clear, implementable requirements; acceptance criteria

3. User Roles and Permissions

3.1 Role Definitions

Four roles are recognised by the system. A single user account may hold multiple roles simultaneously (for example, a user who is both a line manager and an employee).

Role	Scope	Typical User
Admin	Full system access, all employees, all cameras, all configuration	IT administrator
HR	All employees (read), approval queue (final approver), policy configuration, report distribution	HR manager

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Role	Scope	Typical User
Manager	Own assigned department(s) only — team members and their records, approval queue (first approver)	Team lead / department manager
Employee	Own records only — personal attendance, personal request submission	Standard employee

3.2 Role Assignment Rules

- Roles are assigned via the employee Excel import or by Admin within the application. They are not auto-derived from Entra ID group memberships.
- A user may be assigned to multiple departments. When a manager holds multiple departments, they see the union of all employees across those departments.
- A user may simultaneously hold Employee and Manager roles. The dashboard will provide a role switcher allowing them to view the system as either role.
- Admin is not department-scoped; Admin sees all data and can filter as needed.
- HR is a special system role, not a department. A user tagged as HR has approval authority over all departments.

3.3 Permission Matrix

Action	Admin	HR	Manager	Employee
View any employee's attendance	✓	✓	own dept only	self only
Modify employee master data	✓			
Configure shift policies	✓	✓		
Configure holidays / leave policy	✓	✓		
Approve attendance exceptions (1st level)	✓	✓	own dept only	
Approve attendance exceptions (final)	✓	✓		
Submit attendance exception request	✓	✓	✓	✓

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Action	Admin	HR	Manager	Employee
Upload own reference photo	✓	✓	✓	✓ (requires Admin approval)
Import employees from Excel	✓			
Export data (filtered or full)	✓	✓	own dept only	self only
View camera health and statistics	✓			
Manage cameras (add/edit/remove)	✓			
Configure daily report delivery	✓	✓		

4. Functional Requirements

4.1 Authentication and Session Management

FR-AUTH-01 — Microsoft Entra ID (Azure AD) OIDC Login

- The system shall support single sign-on via Microsoft Entra ID using the OIDC protocol.
- The Azure AD tenant, client ID, and client secret shall be configurable by the Admin without code changes.
- User identity shall be matched to the internal employee record using the user's email address returned from Entra ID.
- If a logged-in Entra ID user has no matching employee record in the system, access shall be denied with an explanatory message.

FR-AUTH-02 — Email / Password Login (Development)

- The system shall also support email/password authentication, intended primarily for development and for any user who does not have an Entra ID account.
- Passwords shall be stored using a modern hashing algorithm (bcrypt or Argon2). Plain-text passwords shall never be stored.
- Admin can enable or disable this login method globally. In a production deployment it may be disabled entirely.

FR-AUTH-03 — Session Handling

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- Sessions shall expire after a configurable idle period (default 8 hours).
- Logout shall revoke the session immediately.
- All API access (excluding static assets and login endpoints) shall require a valid session.

4.2 Employee Management

FR-EMP-01 — Employee Import via Excel

- Admin shall be able to import employee master data via an Excel (.xlsx) file.
- The import file shall contain columns for: Employee ID, Name, Email, Department(s), Designation, Role (Admin/HR/Manager/Employee), Manager Email, and any additional custom attributes to be finalised with the client.
- The system shall validate the file before committing: duplicate Employee IDs are rejected; Employee IDs must not contain path-unsafe characters; emails must be unique.
- The system shall provide a dry-run preview showing rows that would be added, updated, or rejected, before the user commits the import.
- The system shall support exporting the current employee list back to Excel for offline editing and re-import.

FR-EMP-02 — Reference Photo Ingestion

- Admin shall be able to upload reference photos by placing files into a designated folder (or dragging them into a bulk uploader in the UI).
- Filename convention: Employee ID is the stem. Examples:
  - *OM0097.jpg — treated as a front-angle reference*
  - *OM0097_front.jpg, OM0097_left.jpg, OM0097_right.jpg — explicit angle tagging*
- The system shall auto-link each photo to the employee whose ID matches the filename stem. No employee record is created automatically — the record must already exist via Excel import.
- Photos for which no employee record exists shall be flagged in a 'pending orphan photos' list visible to Admin.
- The minimum required is 1 photo per employee (front-facing). Multiple photos are recommended for improved off-angle recognition accuracy.
- Admin shall be able to re-run identification processing after adding new photos, so previously unknown events can be retroactively identified.

FR-EMP-03 — Employee Self-Service Photo Upload

- An employee may upload a new reference photo of themselves from their dashboard.
- Uploaded photos shall not be used for identification until approved by an Admin.

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- Until approval, the photo is tagged 'pending' and does not replace the current active photo.

FR-EMP-04 — Multi-Department Assignment

- A user may be assigned to more than one department.
- For a Manager with multiple departments, data access scope is the union of all their departments' employees.

FR-EMP-05 — Search

- Users shall be able to search employees by ID, name, email, or department (text search only).
- Search results are scoped to the user's role permissions (Admin sees all; Manager sees own departments; Employee sees self only).

4.3 Camera and Capture Management

FR-CAM-01 — Camera CRUD

- Admin shall be able to add, edit, enable/disable, and remove IP cameras.
- Each camera has: name, RTSP URL, enabled flag, assigned location/zone (for entry vs. exit tagging), and creation metadata.
- Camera passwords stored in the database shall be encrypted at rest (not plaintext).

FR-CAM-02 — Live Capture

- The system shall support live RTSP capture from the configured cameras.
- Only one camera's live feed is streamed at a time in the UI preview; background capture for additional cameras is a future phase.
- The system shall run face detection in real-time with motion-skip optimisation to minimise CPU when no movement is present.
- Events shall be grouped by person-appearance using the IoU tracker (carried over from existing architecture).
- Up to 10 best face crops per event shall be saved for identification.

FR-CAM-03 — Camera Health Monitoring

- The system shall record, for each camera: current online/offline state, total uptime and downtime, and image capture count.
- Historical health data shall be retained for 30 days.
- This information shall be visible only to Admin.

4.4 Attendance Logic and Time Policies

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FR-ATT-01 — Attendance Record Generation

- For each employee for each working day the system shall derive a daily attendance record.
- 'In time' is the timestamp of the earliest identified event for that employee on that date.
- 'Out time' is the timestamp of the latest identified event for that employee on that date.
- The camera associated with the 'in' event and 'out' event shall both be recorded (they may differ, e.g. Front Door in, Parking out).
- Total hours present shall be computed as (out time − in time), displayed as HH:MM:SS.
- All intermediate detections between in and out shall be preserved in the event log for audit purposes, but are not used for attendance calculation.

FR-ATT-02 — Shift Policy Types

The system supports multiple named shift policies. Admin or HR shall be able to create, edit, and deactivate policies at runtime without a code deployment.

Policy type	Rule	Flags produced
Fixed	In time = 07:30, out time = 15:30 (default; configurable per policy)	'Late in' if in > start; 'Early out' if out < end
Flex	In time between 07:30–08:30; out time between 15:30–16:30 (default)	Must complete 8 hours. If ≥ 8 hrs → no flag (informational 'flex' note only); if < 8 hrs → late-in and/or early-out flags
Ramadan	Admin-configured reduced hours for the Ramadan period	Same late/early logic applied against policy times
Custom / special occasion	Named policy with bespoke in/out times, applied to specific date ranges	Same late/early logic applied against policy times

FR-ATT-03 — Overtime Calculation

- If an employee's total hours on a working day exceeds the policy's defined work duration, the difference shall be recorded as overtime hours on that record.
- Overtime is stored in a separate column/field, not merged into total hours.
- Configurable notification: when a record is flagged with overtime, a notification is sent to the employee's manager and HR (delivery channel: in-app and/or email — configurable).

FR-ATT-04 — Holidays and Leave Policy

- Admin shall be able to upload or configure a holidays list with effective dates.
- On a holiday, no attendance record is expected; missing employees are not flagged as absent; daily reports skip that date's auto-delivery unless explicitly overridden.

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- If an employee is detected on a holiday, their presence shall be recorded and counted as overtime.
- Leave policy (annual leave, sick leave, etc.) shall be configurable. Recorded leaves exempt the employee from 'absent' flags on those dates.

FR-ATT-05 — Policy Assignment

- Shift policies can be assigned per-department, per-employee, or globally. Employee-level overrides beat department-level, which beat global defaults.
- The system shall show, on each attendance record, which policy was applied.

Assumption AS-ATT-01

- When an employee is detected on multiple cameras on the same day, the system treats them as one working day (first in / last out across all cameras). This matches typical CCTV-based attendance. Client to confirm.

4.5 Request and Approval Workflow

FR-APR-01 — Exception Requests

- Any authenticated user shall be able to submit an exception request (e.g., 'I was late because of a doctor's appointment').
- A request may be pre-submitted ('I will be late tomorrow') or retroactive ('I was late yesterday because X').
- Each request has: requester, associated date, reason category (dropdown — finalised with client), free-text notes, optional attachment, status, and audit trail.

FR-APR-02 — Two-Level Sequential Approval

- Submitted requests go first to the employee's line Manager.
- If the Manager approves, the request advances to HR for final approval.
- If the Manager rejects, the request is closed as 'rejected by manager' and does not advance.
- HR can approve or reject. HR decisions are final.
- At every stage, the requester sees the request's current status in their dashboard.
- Notifications (in-app and optionally email) are sent to the next approver and the requester on each state change.

FR-APR-03 — Attendance Display While Pending

- A flagged attendance record with an active request attached is displayed as 'Pending approval' in dashboards and reports.

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- Once approved, the flag is replaced by the approved reason label.
- Once rejected, the original flag ('late in', 'early out', etc.) is restored and marked 'rejected'.

FR-APR-04 — Employee Edit Restriction

- An employee shall not be able to directly edit any attendance record.
- All modifications to an attendance record are applied through the approval workflow.

4.6 Reports and Export

FR-REP-01 — Built-in Report Types

- Event log report — one row per detected appearance. Columns: Camera Name, Date, Time, Identified Person, Reference Photo, Image (embedded thumbnail).
- Daily attendance report — one row per identified person per day. Columns: Camera (In), Employee Name, Employee ID, In Time, Out Time, Date, Total Duration, Overtime, Reference Photo, In Snapshot, Policy Applied, Flags, Pending Request(s).
- Department summary report (new) — aggregated counts per department for a selected date range.

FR-REP-02 — Scheduled Delivery

- Admin or HR shall be able to configure scheduled reports with: report type, recipients (email addresses or distribution lists), schedule (daily / weekly / monthly), delivery method (email attachment or download link).
- Scheduled reports default to daily at 08:00 for the previous day's attendance, sent to HR and IT — configurable by Admin.
- Delivery shall be retried on failure up to 3 times; Admin is notified if delivery ultimately fails.

FR-REP-03 — On-demand Export

- Users shall be able to filter data (date range, department, camera, employee, flag type) and export the filtered set as Excel or PDF.
- Full dataset export is available to Admin and HR only.
- Filter scope respects the user's role permissions.

FR-REP-04 — ERP Integration Hook (future)

- The report generator shall expose a clean interface for downstream systems (ERP) to pull generated reports.
- A specific ERP connector is not in Phase 1 scope. The architectural hook will be included so a future phase can plug in without refactoring.

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4.7 Dashboards

FR-DASH-01 — Admin Dashboard

- Summary cards: total employees enrolled, cameras active, events today, identification rate, pending approvals
- System health: camera status, recent errors, disk usage, log rotation status
- Quick links: add employee, add camera, configure policy

FR-DASH-02 — HR Dashboard

- Company-wide attendance metrics for today and configurable date ranges
- Pending approvals queue (HR-level requests only, meaning Manager-approved)
- Scheduled reports status

FR-DASH-03 — Manager Dashboard

- Own department employees: who's in, who's out, who's late, who has a pending request
- Pending approvals queue (first-level requests from own team)
- Team attendance trend (configurable date range)

FR-DASH-04 — Employee Dashboard

- Today's status (in time, out time if set, total hours, flags, applied policy)
- Monthly calendar view showing own attendance history
- Submit new request button; list of own submitted requests and their statuses
- Upload own photo (pending approval)

FR-DASH-05 — Role Switcher

- When a user holds multiple roles (e.g., Manager + Employee), the dashboard provides a switcher to view the system as either role.

4.8 Filtering and Search

- All major data views (events, attendance, employees) support filtering by: date range, department, camera, employee, role, flag type, request status.
- Filter state is preserved in URL query parameters so filtered views can be bookmarked and shared (permissions permitting).
- Filters respect role permission boundaries.

4.9 Audit Log

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- The system shall maintain a complete audit log of security- and data-relevant events: logins, permission changes, employee record modifications, policy changes, approval decisions, Admin-initiated deletions, bulk imports, scheduled report delivery outcomes.
- Audit log entries are append-only; they cannot be edited or deleted by any role including Admin.
- Audit log viewable by Admin only, with filtering and export.

5. Non-Functional Requirements

5.1 Performance

- Face detection on a single camera feed shall run at an analyzer rate of at least 4 FPS on the target hardware (modern Ubuntu server with 8+ cores).
- API response time for all read endpoints shall be < 500 ms for a dataset of 1,000 employees and 6 months of attendance data.
- Reports (up to 3,000 rows) shall generate in < 30 seconds.
- UI dashboard pages shall render initial content in < 2 seconds on a corporate LAN.

5.2 Availability

- Target availability during office hours: 99.5% (allows ~4 hours/month planned downtime).
- The system shall auto-reconnect to cameras on network blips.
- Critical failures (camera offline > 10 min, database unreachable, disk > 90%) shall surface in the Admin dashboard and optionally notify via email.

5.3 Security

- All HTTP traffic shall be served over HTTPS. Development may use self-signed certificates; production will use a corporate or Let's Encrypt certificate.
- Passwords hashed with bcrypt/Argon2; never stored or logged in plaintext.
- Camera RTSP passwords encrypted at rest in the database using AES-256.
- All user inputs validated server-side; file uploads validated by extension and magic-bytes.
- Role and department scope enforced at the API layer, not only the UI.
- API rate limiting on authentication endpoints to prevent brute-force.
- CSRF protection on state-changing endpoints.
- Data at rest: regular database backups, encryption of sensitive fields.

5.4 Data Protection (Oman PDPL)

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- Personal data (names, emails, face biometrics) handled as 'sensitive personal data' under Oman's Personal Data Protection Law.
- The system shall support: explicit consent recording (where applicable), data-subject access request fulfillment, data deletion on verified request, and a documented retention policy.
- Biometric data (face crops and embeddings) retained indefinitely as per client instruction, with Admin-level manual deletion option. Client acknowledges and accepts this retention approach in writing.
- Specific legal review (including consent wording and PDPL audit) is the client's responsibility.

5.5 Data Retention

Data type	Retention
Employee master records	Retained indefinitely; Admin can delete on request
Face crops (saved JPEGs)	Retained indefinitely; Admin can delete
Face embeddings	Retained indefinitely (tied to face crops)
Attendance records	Retained indefinitely
Approval history	Retained indefinitely (audit requirement)
Audit log	Retained indefinitely
Camera health history	30 days (rolling)
Application logs	30 days with rotation (configurable)

5.6 Backup, Restore, and Disaster Recovery

- The system shall include automated daily backups of: database, face crops folder, reference photos folder, reports folder, configuration.
- Backups shall be written to a configurable local path; off-site backup (e.g. to a NAS or cloud bucket) is a configurable option.

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- A restore script shall be provided, runnable from a single command, that restores the system from a backup archive to a blank Ubuntu host.
- Recovery Point Objective (RPO): 24 hours (last daily backup). Recovery Time Objective (RTO): 4 hours on a standby host.
- A quarterly DR test (documented) is recommended to the client.

5.7 Deployment and Operations

- The system shall be deployable on a single Ubuntu 22.04 LTS host via a single setup script. The script provisions dependencies, configures services, and starts the application.
- Services run under systemd with automatic restart on failure.
- Log files are rotated daily; compressed archives retained per retention policy.
- Health check endpoint /health returns 200 OK with service state for external monitoring.
- Configuration via environment variables and a config file — no hardcoded values.

5.8 Responsive UI

- The web UI shall be fully functional on: desktop (≥ 1280px), tablet (768–1279px), and mobile (< 768px).
- Primary workflows — login, view own attendance, submit request, approve request — must work on mobile.
- Admin configuration screens may prioritise desktop but shall remain usable on tablet.

5.9 Browser Support

- Latest two versions of Chrome, Edge, Safari, and Firefox.
- Internet Explorer is not supported.

5.10 Localisation

- Phase 1: English only.
- All user-facing strings stored in a single translation file to permit future Arabic (RTL) localisation without code rework.

6. Technical Architecture (Summary)

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Full architecture and detailed data model belong in the Technical Design Document that follows this BRD. This section captures the agreed direction only.

6.1 Technology Stack

Layer	Technology	Rationale
Backend API	Python 3.11 + FastAPI	Carries over from existing system; mature, async-capable
Frontend	React 18 + Vite	Carries over from existing system
Database	PostgreSQL 15	Relational schema fits the domain; transactions needed for approval workflow; pgvector extension optional for face similarity
Face detection / embeddings	InsightFace (buffalo_l) on CPU	Carried over; no GPU assumed
Background jobs	Python threading + APScheduler	Simpler than Celery for single-host deployment
Authentication	Authlib (OIDC) + local password for dev	Standards-based, Entra-compatible
Deployment	Ubuntu 22.04 LTS + systemd + Nginx reverse proxy + Let's Encrypt	Single-host model per brief
Containerisation	Optional (Docker Compose)	Optional; setup script will also support bare-metal install

6.2 High-Level Component Diagram (description)

- Nginx (TLS termination, reverse proxy) in front of FastAPI backend and serving the built React frontend
- FastAPI backend — REST API, auth, business logic
- Capture worker — separate thread/process per active camera, reader + analyzer pattern (from existing system)
- Scheduler — runs daily report generation, camera health polls, retention cleanups, backup
- PostgreSQL — primary data store
- File storage — face crops, reference photos, reports (local disk; configurable path)

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- Entra ID (external) — OIDC identity provider

6.3 Data Model (Entities Overview)

The following high-level entities will exist. Field-level detail belongs in the TDD.

- User — identity record; links to Entra ID subject and/or local password
- Employee — master employee data; one-to-one with User
- Department
- EmployeeDepartment — many-to-many association
- Role, UserRole — role assignments
- ShiftPolicy — named policy definitions
- ShiftAssignment — policy assigned to employee/department/globally with effective dates
- Holiday
- LeaveType, LeaveRecord
- Camera
- CameraHealthSnapshot — rolling 30-day history
- Event — one per person-appearance (existing)
- FaceCrop — individual saved face (existing)
- AttendanceRecord — derived per-employee-per-day
- ExceptionRequest, ApprovalStep
- Report, ReportSchedule, ReportDelivery
- AuditLogEntry

7. Assumptions

The following assumptions are made by the engineering team based on the brief and clarifications. Each needs explicit client confirmation during BRD review.

ID	Assumption
A-01	The client will provide a final column list for the employee import Excel template.
A-02	Initial deployment hardware is a single Ubuntu 22.04 host with ≥ 8 cores and ≥ 16 GB RAM. Actual sizing to be confirmed after camera count is finalised.
A-03	All cameras are CP Plus or otherwise RTSP-compatible, accessible over the corporate LAN without NAT or firewall traversal.

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ID	Assumption
A-04	Same-day multi-camera detection counts as a single working day (first in / last out across all cameras).
A-05	Exception-request reason categories are to be finalised with the client during UI design review.
A-06	The notification delivery channel for overtime and workflow events is in-app by default. Email is optional via the Admin's scheduled-report mail account.
A-07	Self-signed HTTPS certificates are acceptable for initial development and internal testing. Production certificate management will be decided before go-live.
A-08	Client will provide Entra ID tenant information (tenant ID, app registration with redirect URI) in advance of the authentication implementation phase.
A-09	Client is responsible for any biometric-data consent wording communicated to employees and for Oman PDPL legal review.
A-10	Arabic UI localisation is not in scope for Phase 1 but will not be architecturally blocked.

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8. Open Questions

The following items were raised during requirements work but not fully answered. They are documented here so they can be resolved before or during the relevant implementation phase.

ID	Question	Needed by
Q-01	Final list of Excel import columns (Employee master) — especially any custom attributes beyond ID, Name, Email, Department, Designation, Role, Manager Email.	Phase: Employee Management
Q-02	Exact list of exception-request reason categories (for the dropdown).	Phase: Approval Workflow
Q-03	Default scheduled-report time and recipients for first deployment.	Phase: Reports
Q-04	Specific ERP product for eventual sync (so the hook can be shaped appropriately).	Phase: Reports (ERP hook)
Q-05	Whether approved HR roles will have separate read-vs-approved permissions, or if all HR see all.	Phase: Roles
Q-06	Camera location/zone taxonomy — how 'entry' vs 'exit' cameras will be tagged.	Phase: Camera Management
Q-07	Whether Admin requires two-factor authentication (recommended).	Phase: Authentication
Q-08	Off-site backup destination (NAS, cloud bucket, both).	Phase: Ops
Q-09	SMTP server for outbound email (reports, notifications) — or preference for Microsoft Graph mail.	Phase: Notifications
Q-10	Approval SLA — how long before a pending request is auto-escalated or auto-closed, if at all.	Phase: Approval Workflow

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9. Phased Delivery Plan

The engineering team proposes delivering the system in seven phases. Each phase produces a demonstrable, reviewable increment. Phases are approximately equal in effort; exact sequencing may be adjusted during technical planning.

Phase	Name	Scope (summary)
P1	Foundation migration	Migrate from SQLite to PostgreSQL; restructure project for multi-user; introduce proper config management; set up backup script skeleton and log rotation.
P2	Authentication & user management	Entra ID OIDC + email/password login; User/Role/Department tables; role-based route protection; basic Admin user-management UI.
P3	Employee import & photo ingestion	Excel import with validation and preview; reference-photo ingestion by filename; orphan-photo management; search.
P4	Shift policies & attendance engine	Shift policy CRUD (Fixed, Flex, Ramadan, custom); holidays and leave policies; daily attendance derivation; overtime and flag computation.
P5	Approval workflow & exception requests	Exception request submission; two-level sequential approval (Manager → HR); dashboards reflecting pending/approved/rejected states; notifications.
P6	Reports & scheduled delivery	Enhanced Daily Attendance report; department summary; scheduled delivery (email, link); on-demand export with filters; ERP hook.
P7	Operations & go-live	Camera health dashboard; audit log UI; setup script, backup/restore/DR scripts; HTTPS configuration; production hardening; documentation.

Each phase concludes with a review checkpoint: demo to the client, acceptance sign-off, and any change requests folded into the next phase's plan before work begins.

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10. Acceptance Criteria (Phase 1 summary level)

Detailed acceptance tests belong in each phase's deliverable. At the project level, the client will accept the system when:

1. Admin can import 50+ employees from Excel, upload their reference photos by filename, and see all of them enrolled correctly.
2. Three user accounts (Admin, Manager, Employee) can each log in via Entra ID and see only the data appropriate to their role.
3. Capture from at least one live camera produces events that are correctly identified against enrolled employees in > 95% of normal-lighting cases.
4. Daily attendance records are computed correctly for Fixed and Flex policies for at least three enrolled employees over one working week.
5. An employee can submit an exception request that routes to their Manager, then HR, with status correctly reflected at every stage.
6. A scheduled daily report delivers successfully to a configured recipient by email for three consecutive days.
7. On-demand export produces a valid Excel file for any filtered view available to the exporting user.
8. The system survives a full restore from backup to a blank Ubuntu host using the provided scripts, with all data intact.
9. All screens are usable on a mobile Safari and Chrome browser.
10. System passes an internal security review covering HTTPS, password hashing, role enforcement, and camera-credential encryption.

11. Sign-off

This Business Requirements Document is presented for stakeholder review. Sign-off is requested on:

- The scope as described in Sections 1.2 and 1.3
- The role and permission model in Section 3
- The functional requirements and attendance logic in Section 4
- The assumptions in Section 7 (each assumption either confirmed or revised)
- The open questions in Section 8 (with commitments to resolve by the phases noted)
- The phased delivery approach in Section 9

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